Data Analytics

Project 4: Data Visualization

Work Report: what we did, step by step, and what we found

Prepared by: **Bilal**
BSIT, Lahore Garrison University (LGU)
DecodeLabs Data Analytics Internship, Batch 2026
Dataset: Orders data (1,200 orders, Jan 2023 - Jun 2025)
Deliverables: Project4_Data_Visualization_Solved.xlsx and this report

1. Project Overview

Project 4 is the optional "Mastery Phase" of the DecodeLabs Data Analytics internship. The goal is to create visual representations of data that communicate insights clearly. The project brief asks for three things: create charts (bar, line, pie etc.), choose the appropriate visual for the data, and highlight key insights. The key skills are data visualization, charts and storytelling with data.

The brief also gives a playbook with three pillars. We followed all three while working:

Pillar	Meaning	How we applied it
1. Choose with purpose (The Architect)	Match chart type to the business question, keep axes honest, no 3D.	Used bars for comparison and a line for time trend. All bars start at zero. No 3D, no pie charts.
2. Edit with precision (The Editor)	Maximise data-ink, remove chartjunk, label directly, use colour as a spotlight.	No gridlines or legends where avoidable, values written on bars, grey for context and one blue accent for the key insight.
3. Tell a story (The Storyteller)	Action titles, SCR framework, one slide one message, answer "So what?".	Every chart has an action title, one message and a recommended action. The final story follows Situation - Complication - Resolution.

2. Dataset Description

We used the orders dataset supplied with the project (Excel file, one sheet). It has **1,200 rows and 14 columns** covering orders from 1 January 2023 to 30 June 2025.

Column	What it contains
OrderID, CustomerID, TrackingNumber	Unique identifiers for the order, the customer and the shipment.
Date	Order date (2023-01-01 to 2025-06-30).
Product	Printer, Tablet, Chair, Laptop, Desk, Monitor, Phone (7 products).
Quantity, UnitPrice, TotalPrice	Units ordered (1-5), price per unit and order value (Quantity x UnitPrice).
ShippingAddress, ItemsInCart	Delivery address and number of items in the cart (1-10).
PaymentMethod	Online, Cash, Credit Card, Debit Card, Gift Card.
OrderStatus	Cancelled, Returned, Pending, Shipped, Delivered.
CouponCode	SAVE10, FREESHIP, WINTER15 (blank when no coupon was used).
ReferralSource	Instagram, Email, Google, Facebook, Referral.

3. Step-by-Step: What We Did

This section lists the work in the order it was done, so it can be used as a record of the project.

Step 1: Read and understood the brief

- Read all 20 pages of the Project 4 training kit (goal, key requirements, chart selection matrix, axis integrity, data-ink ratio, chartjunk, colour rules, action titles, SCR framework, 5-second rule).
- Wrote down the checklist the final work must satisfy (see Section 8).

Step 2: Checked the quality of the data

Check	Result	Action
Missing values	None, except CouponCode (309 blanks)	Blank means "no coupon used", so a helper column Coupon_Clean labels it "No Coupon". Original column kept untouched.
Duplicate rows / duplicate OrderIDs	0 / 0	No rows removed.
Total check: Quantity x UnitPrice = TotalPrice	0 mismatches in 1,200 rows	Totals can be trusted.
Number formatting	Prices had floating-point noise (for example 550.6799999)	Rounded UnitPrice and TotalPrice to 2 decimals.
Date formatting	Dates stored correctly as dates	Formatted as yyyy-mm-dd.

Step 3: Added helper columns in Clean_Data

- **Year** and **Month** (yyyy-mm) from the order date, to build the trend.
- **Coupon_Clean**: coupon code or "No Coupon".
- **Lost_Flag**: 1 if OrderStatus is Cancelled or Returned, else 0.
- **Realised_Revenue**: TotalPrice of orders that are not lost.

Step 4: Built the analysis tables with formulas

In the Analysis sheet we used COUNTIF, SUMIF and SUMIFS so that every number is live and recalculates if the data changes. Tables: by Product, by Order Status, by Referral Source, by Payment Method, by Coupon, H1 comparison by year, and a monthly trend with a 3-month moving average. In total the workbook contains 6,290 formulas and 0 formula errors. All results were also cross-checked independently in Python (pandas) and matched exactly.

Step 5: Chose the right chart for each business question

Business question	Analytical need	Chart chosen
Which products earn the most?	Compare values across categories	Horizontal bar
Is revenue growing or falling?	Trend over time	Line chart (monthly + 3-month average)
How many orders are lost?	Compare counts across statuses	Column chart

Business question	Analytical need	Chart chosen
Which channel brings the most revenue?	Compare values across categories	Column chart
Which payment method loses the fewest orders?	Compare rates across categories	Column chart
Do coupons change order completion?	Compare rates across categories	Column chart

We did not use pie charts, as the brief says to avoid them (and never more than 3 slices).

Step 6: Designed, edited and wrote the story

- Removed gridlines, borders and legends where possible; wrote values directly on the bars.
- Used grey for context and a single blue accent to spotlight the key bar or line; values are also written as numbers so meaning never depends on colour alone.
- Wrote an action title for every chart that states the conclusion.
- Added an insight and a "So what?" action for each chart, and organised the story using SCR.
- Built the KPI cards (top-left of the dashboard) so the main numbers are seen first.

Step 7: Produced the deliverables

- **Excel file** with README, Dashboard, Insights, Analysis and Clean_Data sheets.
- **This PDF report** documenting the work and findings.

4. Key Metrics (KPIs)

Revenue means TotalPrice. The dataset does not state a currency, so values are shown as plain numbers. "Lost" orders are those with status Cancelled or Returned.

KPI	Value	Meaning
Total orders	1,200	All orders from Jan 2023 to Jun 2025.
Gross revenue	1,264,762	Sum of TotalPrice for all orders.
Average order value	1,054	Revenue divided by orders.
Revenue lost (Cancelled + Returned)	519,674	Revenue attached to orders that did not complete (41.1% of the total).
Lost order rate	41.4%	497 of 1,200 orders were Cancelled or Returned.
Realised revenue	745,088	Revenue of orders that are not lost.

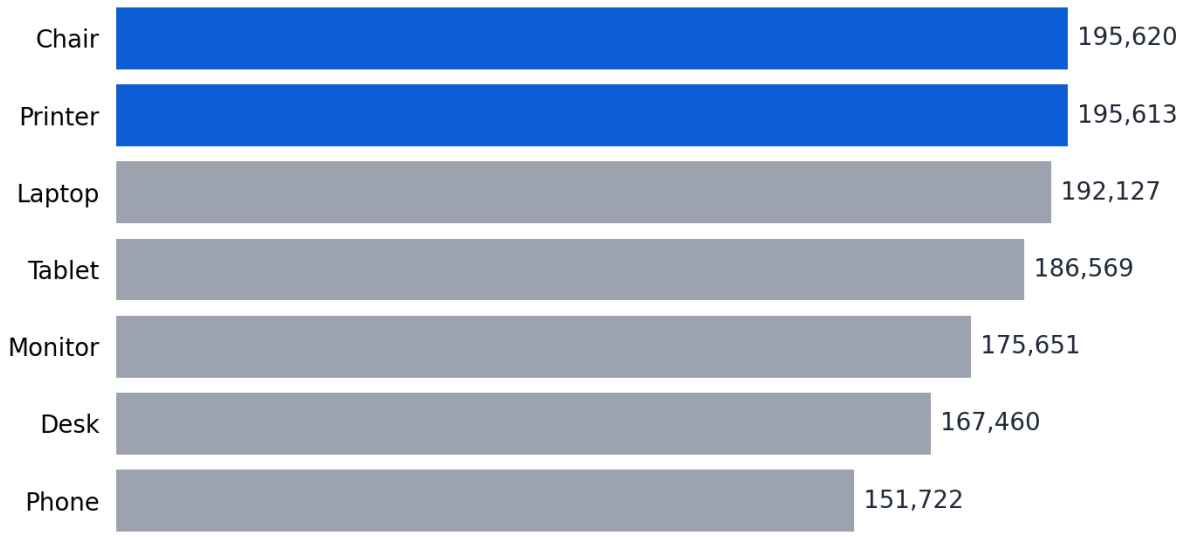
> **Headline** Revenue is falling about 10% a year, and about 4 in every 10 orders never complete. Fixing cancellations and returns is the biggest opportunity in this data.

5. Charts and Insights

Each page below is one slide-style unit: one chart, one message. The title states the conclusion (an action title), followed by the insight and the "So what?".

Chart 5.1

Chairs and Printers lead revenue at about 196K each; Phones trail at 152K



Business question: which products earn the most? Chart: horizontal bar (long category names read best sideways), sorted high to low, starting at zero.

Insight	Chair (195,620) and Printer (195,613) are almost tied at the top. Phone is lowest at 151,722, about 22% below the leaders. The spread is small, with every product contributing between 12% and 15.5% of revenue, so the business does not depend on a single product.
So what?	Keep stock and promotion strong for Chair and Printer. Find out why Phone lags (price, stock, competition) before deciding to reduce it.

Chart 5.2

Revenue is sliding: H1 sales fell from 287K (2023) to 257K (2024) to 232K (2025)



Data Analytics

Project 4: Data Visualization

Work Report: what we did, step by step, and what we found

Prepared by: Bilal

BSIT, Lahore Garrison University (LGU)

DecodeLabs Data Analytics Internship, Batch 2026

Dataset: Orders data (1,200 orders, Jan 2023 - Jun 2025)

Deliverables: Project4_Data_Visualization_Solved.xlsx and this report

1. Project Overview

Project 4 is the optional "Mastery Phase" of the DecodeLabs Data Analytics internship. The goal is to create visual representations of data that communicate insights clearly. The project brief asks for three things: create charts (bar, line, pie etc.), choose the appropriate visual for the data, and highlight key insights. The key skills are data visualization, charts and storytelling with data.

The brief also gives a playbook with three pillars. We followed all three while working:

Pillar	Meaning	How we applied it
1. Choose with purpose (The Architect)	Match chart type to the business question, keep axes honest, no 3D.	Used bars for comparison and a line for time trend. All bars start at zero. No 3D, no pie charts.
2. Edit with precision (The Editor)	Maximise data-ink, remove chartjunk, label directly, use colour as a spotlight.	No gridlines or legends where avoidable, values written on bars, grey for context and one blue accent for the key insight.
3. Tell a story (The Storyteller)	Action titles, SCR framework, one slide one message, answer "So what?".	Every chart has an action title, one message and a recommended action. The final story follows Situation - Complication - Resolution.

2. Dataset Description

We used the orders dataset supplied with the project (Excel file, one sheet). It has **1,200 rows and 14 columns** covering orders from 1 January 2023 to 30 June 2025.

Column	What it contains
OrderID, CustomerID, TrackingNumber	Unique identifiers for the order, the customer and the shipment.
Date	Order date (2023-01-01 to 2025-06-30).
Product	Printer, Tablet, Chair, Laptop, Desk, Monitor, Phone (7 products).
Quantity, UnitPrice, TotalPrice	Units ordered (1-5), price per unit and order value (Quantity x UnitPrice).
ShippingAddress, ItemsInCart	Delivery address and number of items in the cart (1-10).
PaymentMethod	Online, Cash, Credit Card, Debit Card, Gift Card.
OrderStatus	Cancelled, Returned, Pending, Shipped, Delivered.
CouponCode	SAVE10, FREESHIP, WINTER15 (blank when no coupon was used).
ReferralSource	Instagram, Email, Google, Facebook, Referral.

Project 4: Data Visualization | DecodeLabs Data Analytics Internship | Batch 2026 Page 2

3. Step-by-Step: What We Did

This section lists the work in the order it was done, so it can be used as a record of the project.

Step 1: Read and understood the brief

- Read all 20 pages of the Project 4 training kit (goal, key requirements, chart selection matrix, axis integrity, data-ink ratio, chartjunk, colour rules, action titles, SCR framework, 5-second rule).
- Wrote down the checklist the final work must satisfy (see Section 8).

Step 2: Checked the quality of the data

Check	Result	Action
Missing values	None, except CouponCode (309 blanks)	Blank means "no coupon used", so a helper column Coupon_Clean labels it "No Coupon". Original column kept untouched.
Duplicate rows / duplicate OrderIDs	0 / 0	No rows removed.
Total check: Quantity x UnitPrice = TotalPrice	0 mismatches in 1,200 rows	Totals can be trusted.
Number formatting	Prices had floating-point noise (for example 550.6799999)	Rounded UnitPrice and TotalPrice to 2 decimals.
Date formatting	Dates stored correctly as dates	Formatted as yyyy-mm-dd.

Step 3: Added helper columns in Clean_Data

- **Year** and **Month** (yyyy-mm) from the order date, to build the trend.
- **Coupon_Clean**: coupon code or "No Coupon".
- **Lost_Flag**: 1 if OrderStatus is Cancelled or Returned, else 0.
- **Realised_Revenue**: TotalPrice of orders that are not lost.

Step 4: Built the analysis tables with formulas

In the Analysis sheet we used COUNTIF, SUMIF and SUMIFS so that every number is live and recalculates if the data changes. Tables: by Product, by Order Status, by Referral Source, by Payment Method, by Coupon, H1 comparison by year, and a monthly trend with a 3-month moving average. In total the workbook contains 6,290 formulas and 0 formula errors. All results were also cross-checked independently in Python (pandas) and matched exactly.

Step 5: Chose the right chart for each business question

Business question	Analytical need	Chart chosen
Which products earn the most?	Compare values across categories	Horizontal bar
Is revenue growing or falling?	Trend over time	Line chart (monthly + 3-month average)
How many orders are lost?	Compare counts across statuses	Column chart

Project 4: Data Visualization | DecodeLabs Data Analytics Internship | Batch 2026 Page 3

Business question	Analytical need	Chart chosen
Which channel brings the most revenue?	Compare values across categories	Column chart
Which payment method loses the fewest orders?	Compare rates across categories	Column chart
Do coupons change order completion?	Compare rates across categories	Column chart

We did not use pie charts, as the brief says to avoid them (and never more than 3 slices).

Step 6: Designed, edited and wrote the story

- Removed gridlines, borders and legends where possible; wrote values directly on the bars.
- Used grey for context and a single blue accent to spotlight the key bar or line; values are also written as numbers so meaning never depends on colour alone.
- Wrote an action title for every chart that states the conclusion.
- Added an insight and a "So what?" action for each chart, and organised the story using SCR.
- Built the KPI cards (top-left of the dashboard) so the main numbers are seen first.

Step 7: Produced the deliverables

- **Excel file** with README, Dashboard, Insights, Analysis and Clean_Data sheets.
- **This PDF report** documenting the work and findings.

4. Key Metrics (KPIs)

Revenue means TotalPrice. The dataset does not state a currency, so values are shown as plain numbers. "Lost" orders are those with status Cancelled or Returned.

KPI	Value	Meaning
Total orders	1,200	All orders from Jan 2023 to Jun 2025.
Gross revenue	1,264,782	Sum of TotalPrice for all orders.
Average order value	1,054	Revenue divided by orders.
Revenue lost (Cancelled + Returned)	519,674	Revenue attached to orders that did not complete (41.1% of the total).
Lost order rate	41.4%	497 of 1,200 orders were Cancelled or Returned.
Realised revenue	745,098	Revenue of orders that are not lost.

Headline: Revenue is falling about 10% a year, and about 4 in every 10 orders never complete. Fixing cancellations and returns is the biggest opportunity in this data.

Project 4: Data Visualization | DecodeLabs Data Analytics Internship | Batch 2026 Page 4

Business question: is revenue growing or falling? Chart: line chart (best for a trend over time). Grey = monthly revenue, blue = 3-month average. Axis starts at zero.

Insight

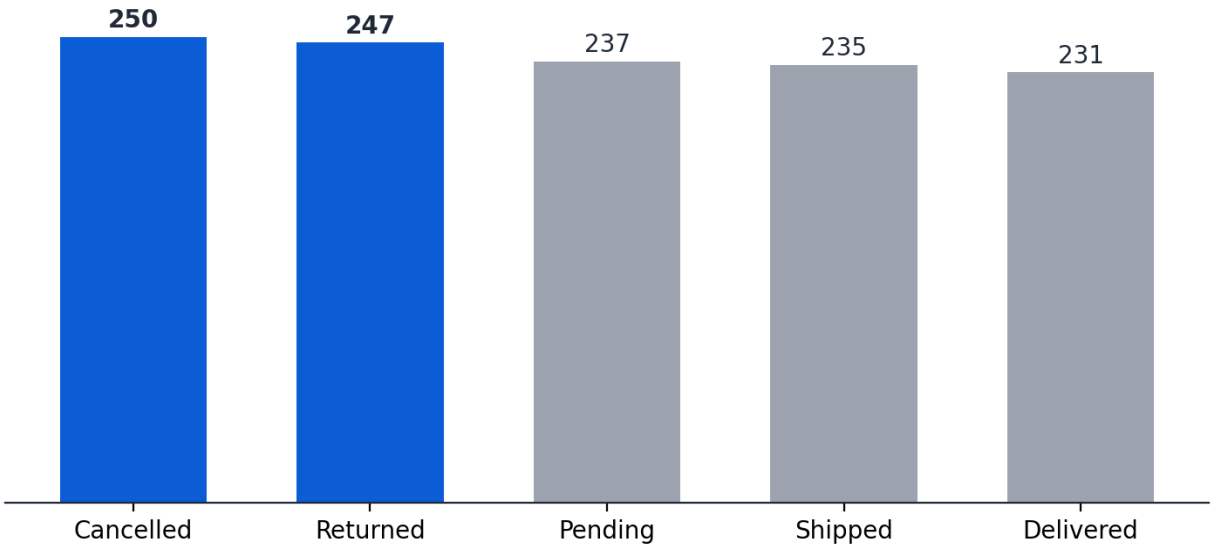
Monthly revenue is noisy, but the 3-month average shows a clear downward drift. Comparing the same half-year (Jan-Jun) keeps the comparison fair: 286.5K, then 257.1K (-10.3%), then 231.9K (-9.8%). Across the three years June is the strongest month (170.6K) and September the weakest (69.3K).

So what?

Plan campaigns ahead of the June peak, and run a recovery campaign for the weaker Jul-Dec period.

Chart 5.3

Cancelled and Returned orders (497) outnumber Delivered ones (231) more than 2 to 1

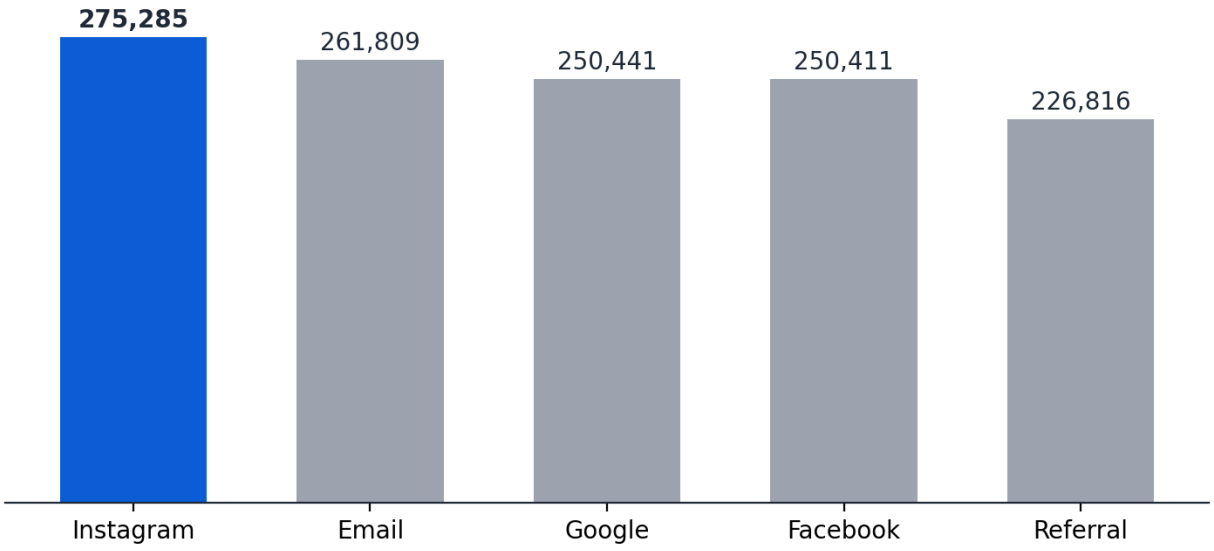


Business question: how many orders are lost? Chart: column chart of order counts by status. Blue = the two loss statuses.

Insight	Cancelled (250) and Returned (247) are the two biggest statuses. Only 231 orders (19%) are Delivered. Cancelled plus Returned orders carry 519.7K of revenue, 41% of the total.
So what?	Top priority. Collect cancellation and return reasons, improve product information and checkout, and track the lost-order rate as a monthly KPI.

Chart 5.4

Instagram is the top traffic source: 275K revenue and the fewest lost orders

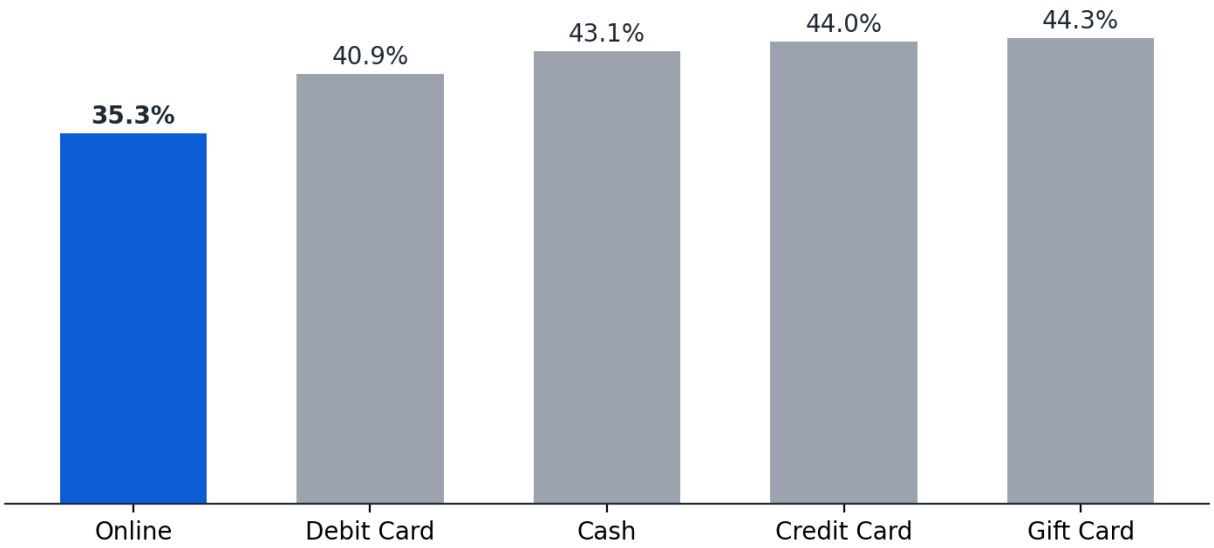


Business question: which channel brings the most revenue? Chart: column chart sorted high to low, starting at zero.

Insight	Instagram brings 275,285 revenue, the most of the five channels, and also has the lowest loss rate (37.1%). Email (43.2%) and Facebook (43.0%) lose more of their orders. Referral is the smallest channel at 226,816.
So what?	Move more marketing budget toward Instagram, and review the targeting of Email and Facebook campaigns that bring orders which later cancel.

Chart 5.5

Online payments lose the fewest orders (35%) vs Gift Cards (44%)

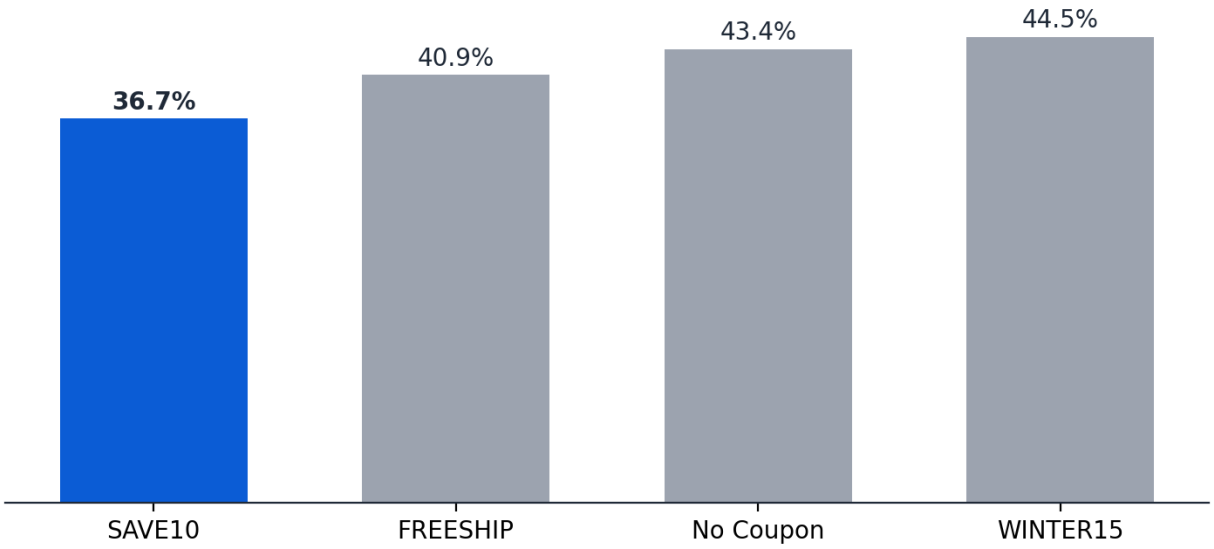


Business question: which payment method loses the fewest orders? Chart: column chart of the loss rate (lost orders divided by orders), starting at zero.

Insight	Online payment has the lowest loss rate at 35.3%. Gift Card is highest at 44.3% and Credit Card is close at 44.0%. Debit Card is 40.9% and Cash 43.1%.
So what?	Encourage online payment at checkout, and review the refund and return process for Gift Card and Credit Card orders.

Chart 5.6

SAVE10 customers cancel or return least (37%); WINTER15 the most (45%)



Business question: do coupons change order completion? Chart: column chart of the loss rate by coupon, starting at zero.

Insight	SAVE10 has a 36.7% loss rate, FREESHIP 40.9%, No Coupon 43.4% and WINTER15 44.5%. Average order value is almost the same across all coupons (about 1,036 to 1,070), so coupons are not changing basket size.
So what?	Prefer SAVE10-style offers and reconsider WINTER15, which does not improve order completion.

Caution: the differences between payment methods, coupons and channels are modest (a few percentage points on 220 to 310 orders each). They are signals worth testing with experiments, not proof of cause.

6. The Data Story (SCR Framework)

Part	Story
Situation	Over 30 months the business took 1,200 orders worth 1.26M in gross revenue, about 1,054 per order, spread evenly across 7 products, 5 marketing channels and 5 payment methods.
Complication	Revenue is falling: H1 revenue dropped from 286.5K (2023) to 257.1K (2024) to 231.9K (2025), about -10% a year. At the same time 41% of orders (497 of 1,200) end as Cancelled or Returned, putting 519.7K of revenue at risk. Only 19% of orders are Delivered.
Resolution	Keep the orders we already win: reduce cancellations and returns first, push the channels and options that lose fewer orders (Instagram, Online payment, SAVE10), and run a recovery plan for the declining trend.

7. Recommended Next Steps ("So what?")

Priority	Action	Why (evidence)
1	Find the reasons for cancellations and returns and fix the top causes.	41% of orders and 519.7K revenue are lost.
2	Shift marketing budget toward Instagram; audit Email and Facebook targeting.	Instagram: highest revenue and lowest loss rate (37.1%).
3	Promote online payment; review Gift Card and Credit Card refund process.	Online loses 35.3% vs about 44% for Gift and Credit Card.
4	Keep SAVE10-style offers; reconsider WINTER15.	SAVE10 loss rate 36.7% vs WINTER15 44.5%.
5	Plan a recovery campaign and time promotions for the June peak.	H1 revenue is down about 10% each year; June is the strongest month.
6	Protect Chair and Printer stock; review why Phone lags.	Top two products earn about 196K each, Phone 152K.

8. Requirements Checklist

Requirement from the project brief	Done	Where
Create charts (bar, line, pie, etc.)	Yes - bar, column and line charts (pie avoided on purpose)	Dashboard sheet; Section 5
Choose appropriate visuals for the data	Yes - chart picked from the business question	Section 3, Step 5
Highlight key insights	Yes - blue accent, action titles, insight text	Dashboard, Insights; Section 5
Axis integrity (bars start at zero)	Yes	All charts
No 3D effects, no chartjunk	Yes	All charts
Direct labelling instead of legends	Yes (the line chart keeps a small legend for its two lines)	All charts
Colour as a spotlight (5-7 hues max, redundant encoding)	Yes - grey plus one blue; values also written as numbers	All charts
Action titles that state the conclusion	Yes	Every chart
SCR storytelling and "So what?"	Yes	Sections 6 and 7; Insights sheet
KPIs at the top-left	Yes	Dashboard KPI cards

9. Guide to the Excel File

Sheet	Content
README	Goal, sheet guide, data checks, definitions, assumptions, design rules.
Dashboard	6 KPI cards and 6 charts with action titles.
Insights	SCR story, one insight and "So what?" per chart, caution note.
Analysis	All summary tables built with live formulas (COUNTIF, SUMIF, SUMIFS, AVERAGE).
Clean_Data	1,200 orders: the original 14 columns plus 5 helper columns (grey headers).

10. Tools Used and Conclusion

Tools: Microsoft Excel formulas and native Excel charts for the workbook; Python (pandas) to double-check every number; Python (matplotlib and ReportLab) for the chart images and this report.

Conclusion: Turning the raw orders data into charts showed a clear story. Revenue has declined by about 10% in each of the last two half-years, and roughly 4 in 10 orders end as Cancelled or Returned. Instagram, online payment and the SAVE10 coupon are associated with fewer lost orders and are the best places to start. The project shows how a chart chosen for the question, kept clean and given an action title, can move a reader from numbers to a decision.

Limitations: the dataset has no currency or cost information, so profit cannot be measured; the dataset has no cancellation reasons, so the causes of lost orders are still unknown; and the differences between categories are small enough that they should be confirmed with further testing.